Final Exam Slides

ECE 3011



Georgia Tech College of Engineering
School of Electrical
and Computer Engineering

Updated: Spring 2023

Guidance

- The following 5 slides are what you will use to present for the Final Exam
  - Do not add slides
    - The format of most of these slides are quad-slides
    - This is the expected format in most business communication settings
  - Focus on non-technical communication
  - Consider what your audience would want to see
    - Imagine that your audience is non-technical investors
    - Remember a picture is worth 1000 words

Customer Discovery and Define

- Team 02
  - Christian Santiago
  - Victor Alfaro
  - Sangeon Jeon

Problem Space

- Customer
  - College Students
- Problem
  - College students often go to sleep late and have trouble with waking up in the morning consistently.

Interviews

How many people did you talk to?

4 (Victor)+ 3 (Sangeon) + 6(Christian)

What did you learn?

- For a person to invest in a new alarm clock, a common statement was that the alarm clock must have an “Annoyance factor” to it.
- What are the themes?
- A waking up method that cannot be ignored.

What are the underlying needs?

- A weekly calendar app to take wake-up times as inputs through iPhone "widgets" for ease of use or just in app changes for android and other devices. Communicates directly to the ring via Bluetooth.
- Adjusting the intensity of the alarm clock based on in-app declaration of how important it is that they must be awake. Stronger more pulse-like vibrations that have a rougher wake effect.

Thesis

X will buy Y because of Z

- X – has to be a collection of people (not a hospital)
  College Students
  - 4 (Victor) + 3 (Sangeon) + 6 (Christian)
- Y – DO NOT DEFINE THIS YET!
- Z – Why will they buy or use your product/service
  1. It provides a sure fire way to wake up without the possibility of sleeping in.
  2. Modification of the intensity of the alarm clock.
     (One can have a sure way of estimating when they will need to wake up)

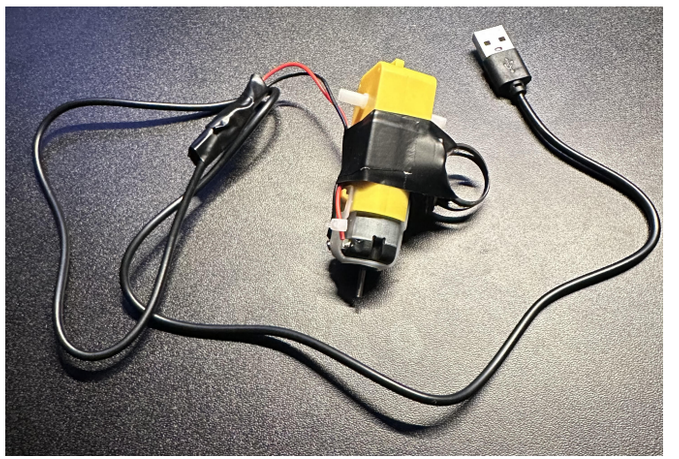
Ideate and Prototype

Thesis

X will buy Y because of Z

- X – College Students struggling with irregular sleep patterns and deep sleep
- Y – "Ring of Alarm" device that delivers haptic stimulus for awakening
- Z – Enhanced reliability and effectiveness in ensuring timely and efficient wake-up

Ideate Process



Hypotheses Statements

H1. Many students have difficulty waking up to their cell phone alarms.
H2. These students are typically willing to invest $30 in a solution to this problem.
H3. Students favor portable alarm clocks over non-portable ones.
H4. Most students are open to the idea of wearing a ring as big as a "class ring" as an alarm device, with a length of 1.3cm and diameter of 2.3cm.
H5. People would prefer a ring over wearables such as bracelets, necklaces, and anklets.
H6. Given the choice between a portable option (requiring battery charging and having reminder functionality) and a non-portable option (powered by an outlet and only providing a morning wake-up alarm), over 90% of the students prefer the portable option.

Ugly MVP

- Haptic alarm clock will fit snug around users' finger with various lengths that vary in max intensity depending on user needs.
- Small LED indicators to display if alarm is active/inactive and live indicator for phone connection.
- Vibrational (Haptic) sections are on the ring that are the main stimulus to awaken user.
- Ring can sync with a companion mobile app where the alarms you set can be integrated and set with varying intensities along with timers.

Test Results (26 Respondents/Interviews)

H1. Many students have difficulty waking up to their cell phone alarms.

- **Majority of students(~63%, Am)** interviewed have **difficulty waking up to cell phone alarms** and often use other alternatives for a better chance of waking up on time.

H2. These students are typically willing to invest $30 in a solution to this problem.

- Of the American students interviewed the **majority (~80%, Am)** said they would invest the **$30**
- Of the Korean students interviewed the **majority (7/10, Kor)** said they would invest the **$10**

Given that (1) The app was well made (2) Others can vouch that it works (3) Charge would last throughout an entire day

H3. Students favor portable alarm clocks over non-portable ones.

- Of the **American students interviewed ~90% desired portability** over hard-wired alarm clocks as it gave way for "naps on campus" and "reminders of class sessions" with emphasis on the application of the "pomodoro technique".
- Korean students**(8/10, Kor) disliked the idea of charging alarm clock daily**. **Battery duration** should be the primary concern.

H4. Most students are open to the idea of wearing a ring as big as a "class ring" as an alarm device, with a length of 1.3cm and diameter of 2.3cm.

- Using Christian's ring, the diameter served well, the space between the skin and the top of the ring served well, and an extension of overall length was a comfortable fit according to most interviewees.

H5. People would prefer a ring over wearables such as bracelets, necklaces, and anklets.

- Almost everyone agreed with this which might have needed a deeper revision or was an oversight on our part. Reasons included
  - Necklaces can vibrate on pillow instead of skin defeating the purpose
  - Anklets to some felt uncomfortable to possess and made points that the vibrations could pose a mildly uncomfortable feeling as well
  - Bracelets could work but a ring would be a lot less impeding with movement.
- People dislike to wear Apple Watch during a sleep(6/10, Kor) **whereas a ring is more much more acceptable (3/10, Kor).**

H(extra1). People are more prone to wake if a stimulus is sent through the finger.

- Using the ugly MVP of a USB wired motor to a ring the interviewees stated that the vibration of the motor could be faster than the motor we had on hand, but it would wake them up with the motor sensation of the ugly mvp. People compared the feeling to a phone ringing for a call, almost impossible to miss and garners their attention.

H(extra2). People prefer portable alarm controlled by app over the alarm with the interface on itself.

- **More than half of the students(8/10, Kor)** preferred app control.

Georgia Tech
CREATING THE NEXT

Final Idea Pitch – One slide project summary of MVP

Background

- College students
- From inspection, we know a majority of students...
  - Use "auditory" alarm clocks
  - Have prior experience with waking up unreliably to these alarms

Pre-existing	Ours
Auditory	Haptic
Outlet + Morning Only	Battery + Reminder
Apple Watch	Ring of Alarm

Solution

- An innovative haptic ring that employs vibrations, gentle electric shocks, and subtle poking sensations to ensure a more reliable and effective wake-up experience. This wearable offers a touch-based approach to waking up that is hard to ignore!

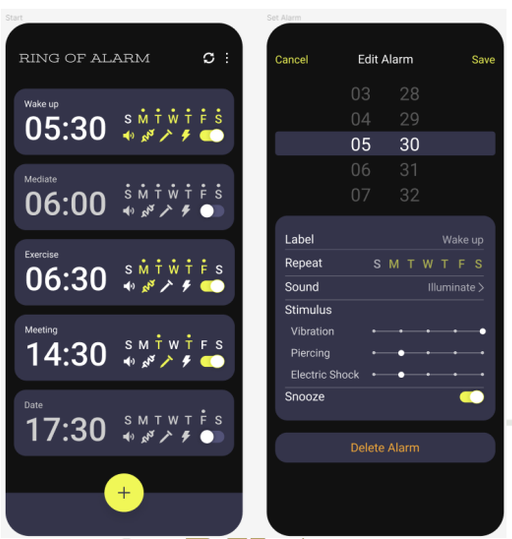


Testing and Market Survey

- A ring alarm can make enough stimulus to wake one up (Tested with Ugly MVP)
- A portable reminder alarm is preferred over a hard-wired one with the battery problem set aside.
- Among the accessories, a ring is the most acceptable to wear during sleep. (Superior to Apple Watch)
- People prefer app control over an alarm with the interface on itself.
- Controllable vibrations allow for versatility in user sensitivity to stimulus

MVP Picture





and Computer Engineering

Conclusion

Go/No Go Decision

GO!!

- Most of the hypotheses turned out to be true.
- If we could find a way to reduce the burden of charging for users, our product might be able to compete with even the Apple Watch in terms of wake-up alarm services.

Surprises

- We did not expect people to favor the ring stimulus over a blanket stimulus.
- 30$ was a good price point if there was a good word of mouth.
- A ring was the preferred device. One would think a necklace since that seems to be the most worn fashion accessory.
- Few people experience embarrassing moments when their alarm rings at an inappropriate moment. I thought it would be one of the main reasons for buying our product.

Class Feedback

- The class was overall structured very well, however some knowledge checks seemed too in depth such as the Ethics video one.
- The requirements for the robot project would be better to be clearly stated in a form of pdf rather than ppt.
- More information about the available material should be provided.
  - MicroBit uses block coding in consideration to teach kids about computer science.
  - Eagle has quite strict dimension limits whereas the other EDAs do not.
- Benjamin Yang should teach ECE 3011 for the next several decades as it was a great experience.

Test Results (26 Respondents/Interviews)

H1. Many students have difficulty waking up to their cell phone alarms.
- **Majority of students(~63%, Am)** interviewed have **difficulty waking up to cell phone alarms** and often use other alternatives for a better chance of waking up on time.

H2. These students are typically willing to invest $30 in a solution to this problem.
- Of the American students interviewed the **majority (~80%, Am)** said they would invest the **$30**
- Of the Korean students interviewed the **majority (7/10, Kor)** said they would invest the **$10**

Given that (1) The app was well made (2) Others can vouch that it works (3) Charge would last throughout an entire day

H3. Students favor portable alarm clocks over non-portable ones.
- Of the **American students interviewed ~90% desired portability** over hard-wired alarm clocks as it gave way for "naps on campus" and "reminders of class sessions" with emphasis on the application of the "pomodoro technique".
- Korean students**(8/10, Kor) disliked the idea of charging alarm clock daily**. **Battery duration** should be the primary concern.

H4. Most students are open to the idea of wearing a ring as big as a "class ring" as an alarm device, with a length of 1.3cm and diameter of 2.3cm.
- Using Christian's ring, the diameter served well, the space between the skin and the top of the ring served well, and an extension of overall length was a comfortable fit according to most interviewees.

H5. People would prefer a ring over wearables such as bracelets, necklaces, and anklets.
- Almost everyone agreed with this which might have needed a deeper revision or was an oversight on our part. Reasons included
  - Necklaces can vibrate on pillow instead of skin defeating the purpose
  - Anklets to some felt uncomfortable to possess and made points that the vibrations could pose a mildly uncomfortable feeling as well
  - Bracelets could work but a ring would be a lot less impeding with movement.
- People dislike to wear Apple Watch during a sleep(6/10, Kor) **whereas a ring is more much more acceptable (3/10, Kor).**

H(extra1). People are more prone to wake if a stimulus is sent through the finger.
- Using the ugly MVP of a USB wired motor to a ring the interviewees stated that the vibration of the motor could be faster than the motor we had on hand, but it would wake them up with the motor sensation of the ugly mvp. People compared the feeling to a phone ringing for a call, almost impossible to miss and garners their attention.

H(extra2). People prefer portable alarm controlled by app over the alarm with the interface on itself.
- **More than half of the students(8/10, Kor)** preferred app control.

Georgia Tech
CREATING THE NEXT

- Use this space to present your Hypothesis and Ugly MVP test results
- Consider non-technical communications
- What is the most important information
- How do you convey the most important information using limited space